

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Eastern Wood-Pewee"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

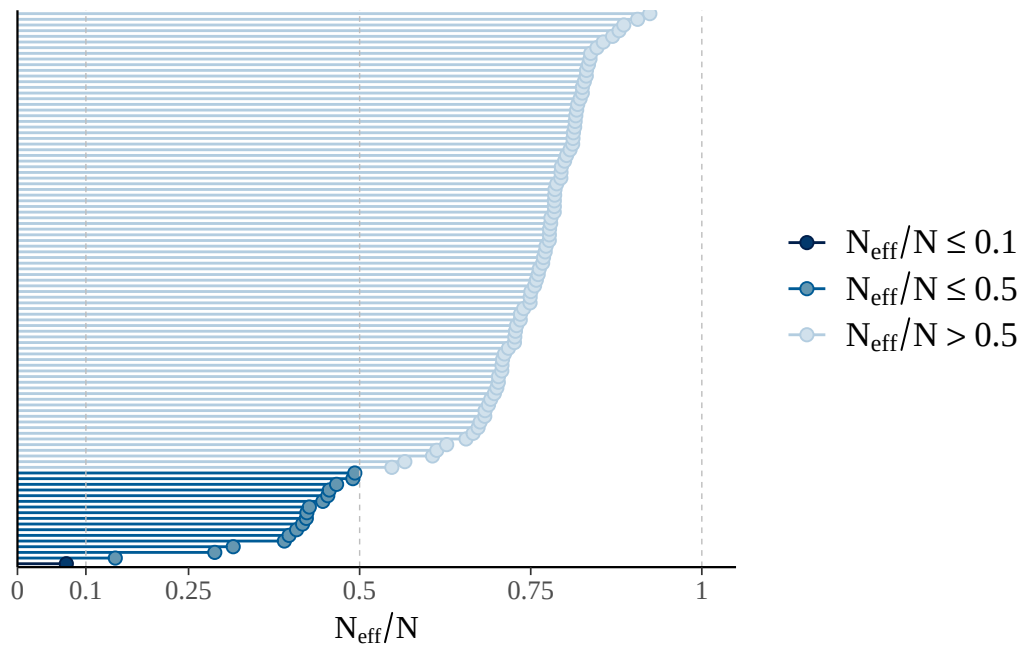
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size too small"
```

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.1917	0.9377	1.1485	1.1594	1.3910	2.5075

```
[1] "n_eff ratio tail ESS:"
```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
	0.07141	0.69953	0.76165	0.73903	0.81187	0.92401

```
[1] "effective sample size in bulk or tail too small"
```

Do chains mix well:

No divergences to plot.

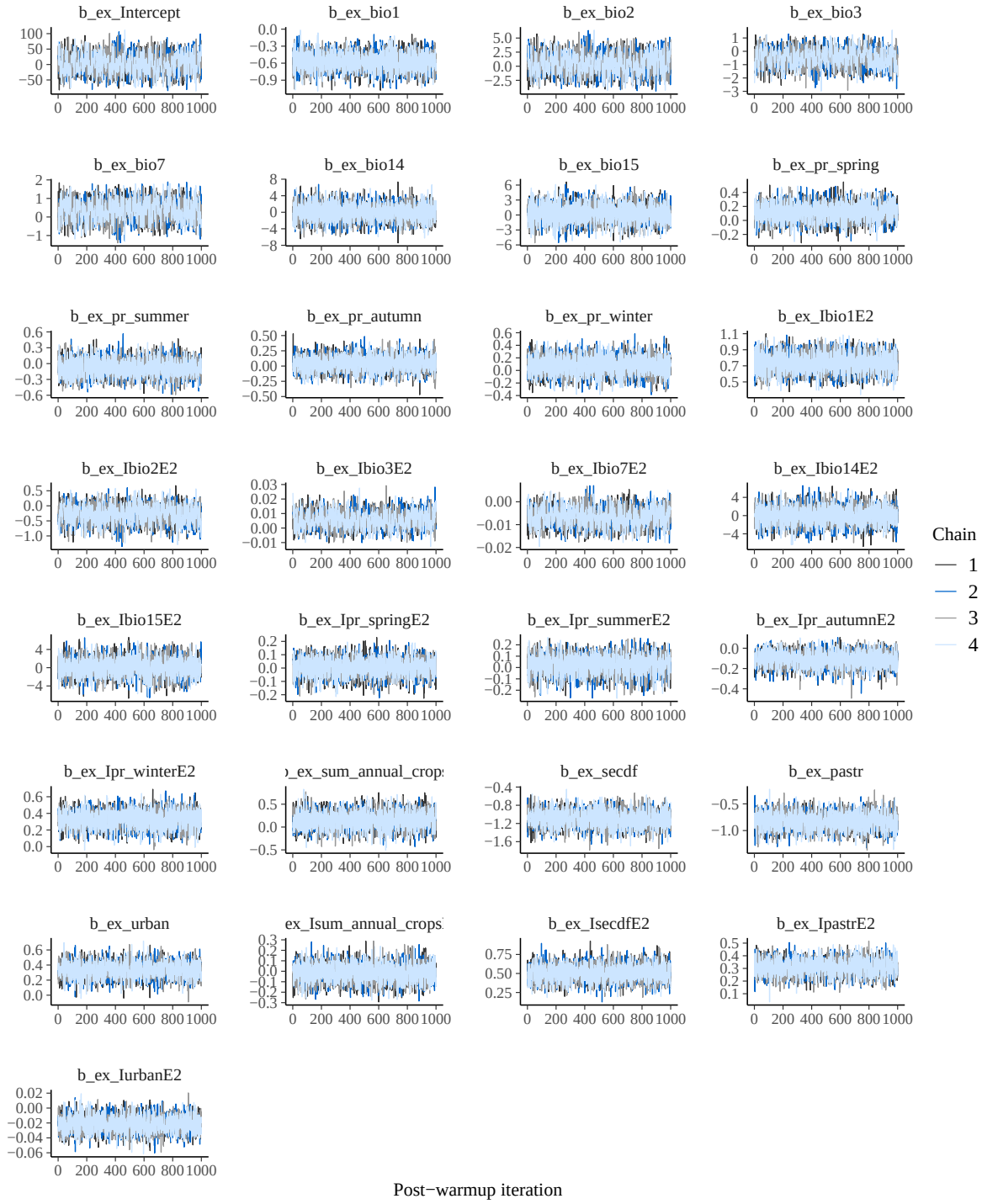
No divergences to plot.



No divergences to plot.

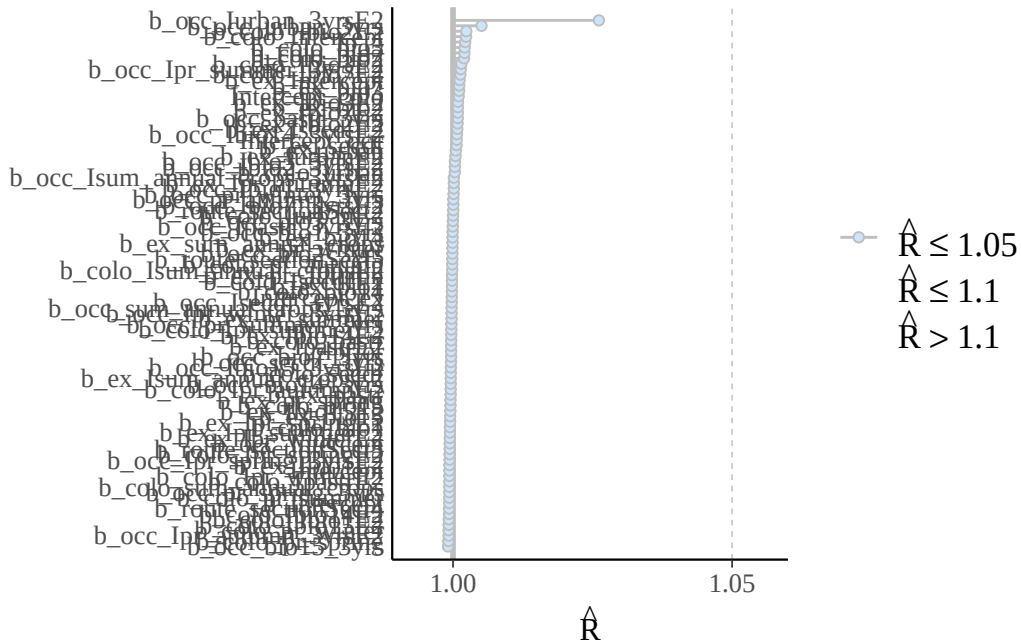


Post-warmup iteration



Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9991	0.9995	0.9998	1.0004	1.0005	1.0261



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"      "b_occ_bio1_3yrs"
[3] "b_occ_Ibio1_3yrsE2"   "b_occ_Ipr_summer_3yrsE2"
[5] "b_occ_secdf_3yrs"     "b_occ_Isecdf_3yrsE2"
```

```
[1] "b_colo_bio3"           "b_colo_pr_autumn"
[3] "b_colo_lbio2E2"        "b_colo_lpr_summerE2"
[5] "b_colo_lpr_autumnE2"   "b_colo_sum_annual_crops"
[7] "b_colo_secdf"          "b_colo_pastr"
[9] "b_colo_urban"          "b_colo_lsecdfE2"
[11] "b_colo_lurbanE2"
```

[1]	"b_ex_bio1"	"b_ex_Ibio1E2"	"b_ex_Ipr_winterE2"
[4]	"b_ex_secdf"	"b_ex_pastr"	"b_ex_urban"
[7]	"b_ex_IsecdfE2"	"b_ex_IpastrE2"	

